

Second Terminal Examination - 2079

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [$11 \times 1 = 11$]

- The resultant of two forces of equal magnitudes, acting at right angle to each other is 100N. Then the magnitude of each force is
 - 70.7N
 - 50N
 - 141.4N
 - 100N
- Which of the following is odd one?
 - electric field intensity
 - force
 - momentum
 - potential
- The number of significant figures in 0.1020 are
 - 4
 - 5
 - 3
 - 2
- A stone is dropped from top of a tower. If it reaches the earth in 6 seconds, the height of tower is nearly
 - 360m
 - 180m
 - 100m
 - 60m
- To a person running with a velocity of 4km/hr, the rain appears to fall to him with a velocity of 5km/hr. The actual velocity of rain is
 - 7km/hr
 - 5km/hr
 - 3km/hr
 - 1km/hr

6. Amount of heat required to raise the temperature of a body through 1K is called its
- a) specific heat capacity b) thermal heat capacity
c) water equivalent d) none
7. If the object is real, a convex mirror always forms
- a) Real and inverted image b) Real and erect image
c) Virtual and erect image d) Virtual and inverted image
8. The angle of incidence for grazing incidence in prism is :
- a) 0° b) 45° c) 60° d) 90°
9. The relation for angle of incidence at minimum angle of deviation is :
- a) $i = \frac{A}{2}$ b) $i = A + \delta_m$
c) $i = \frac{A + \delta_m}{2}$ d) $i = \frac{A - \delta_m}{2}$
10. The refractive index of glass is 1.5 and water is 1.33. Then the critical angle for glass water interface is
- a) 54.6° b) 48° c) 62.73° d) 56°
11. The distance between two charges is increased by 25% then the percentage change in force is
- a) 36% b) 56% c) 65% d) 50%

Group – B

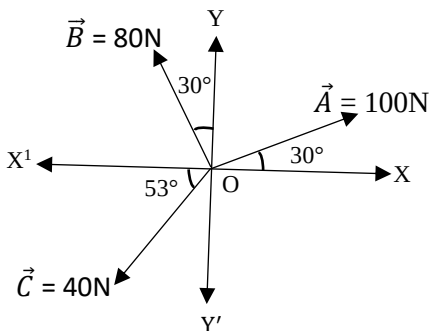
Give short answer to the following questions.

[8 × 5 = 40]

12. a. If $y = a + bt + ct^2$, where y is distance and t is time. What is the dimension and units of b and c? [2]
b. What is the difference between accurate and precise measurement? [1]

- c. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} - \vec{B} = \vec{C}$ and $A - B = C$. Find the angle between $\vec{A}$ and $\vec{B}$. [2]

13. a. Distinguish between average velocity and instantaneous velocity. [1]
 b. Three horizontal ropes pull on a large stone stuck into the ground producing the forces $\vec{A}$, $\vec{B}$ and $\vec{C}$ as shown in the figure.



Using the method of components, find the magnitude and direction of resultant force on the stone. [3+1]

14. a. State the parallelogram law of vector addition. Using this law, find the magnitude and direction of resultant of two vectors $\vec{A}$ and $\vec{B}$ inclined at angle ' θ ' with each other. [3]
 b. Velocities of two bodies A and B are given in vector notation as $\vec{v}_A = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{v}_B = 3\hat{i} + 2\hat{j} - \hat{k}$ respectively. What will be the relative velocity of B with respect to A in this notation? [2]
15. a. Briefly explain the concept of thermal equilibrium and temperature. [2]
 b. Using the following data, determine the temperature at which wood will just sink in benzene. [3]

Density of benzene at $0^\circ \text{C} = 9.0 \times 10^2 \text{ kg m}^{-3}$

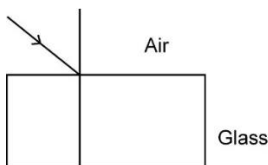
Density of wood at $0^\circ \text{C} = 8.8 \times 10^2 \text{ kg m}^{-3}$

Cubical expansivity of benzene $= 1.2 \times 10^{-3} \text{ K}^{-1}$.

Cubical expansivity of wood $= 1.5 \times 10^{-4} \text{ K}^{-1}$.

16. a. The temperature of metal ball is raised. Which of the following will undergo the largest percentage out of mass, circumference, surface area and volume. increase and which one the least. [2]
- b. We can use a bimetallic strip made of brass and steel each of different length whose length would change in such a way that difference between their lengths remain 10 cm for all temperature. What should be length of each strip ? [Given, linear expansivity of brass = $18 \times 10^{-6} / ^\circ\text{C}$ and Steel = $12 \times 10^{-6} / ^\circ\text{C}$] [3]
17. A pencil is placed 6 cm in front of a concave mirror having a radius of curvature of 40 cm.
- a. What is its focal length? [1]
- b. Determine the position of the image formed by this mirror. [2]
- c. Draw a ray diagram and verify your previous results. [2]

18. a. Complete the path of the ray in the figure shown to show lateral shift.[1]



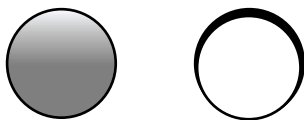
- b. Write the expression of lateral shift as seen in the figure above. [1]
- c. Sketch a graph to explain how the lateral shift varies with angle of incidence. [2]
- d. For what angle of incidence, the lateral shift produced by a parallel sides glass slab is zero? [1]
19. a. What do you mean by quantization of charge? Explain. [2]
- b. James, during his visit to physics lab, sees a neutral conductor. He wishes to make it positively charged. Explain with figure, the steps to be followed to make the body charged positively by method of induction. [3]

Group – C

Give long answer to the following questions.

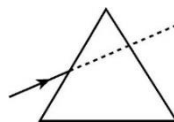
[3 × 8 = 24]

20. a. Show that path followed by projectile fired at angle of 'θ' with a horizontal is parabola and derive the general expression for time of flight and horizontal range. [4]
- b. A river 600m wide flows at the rate of 5km/hr. A swimmer who can swim at the rate of 10km/hr on still water, wishes to cross the river straight.
- i. Along what direction must he strike? [2]
- ii. How much time would he take to cross river? [1]
- c. What does the slope of position-time graph indicate? [1]
21. a. A metal cube is heated from $\theta_1^\circ \text{C}$ to $\theta_2^\circ \text{C}$ so that length increases from l_1 to l_2 and volume from V_1 to V_2 as
- $$l_2 = l_1 [1 + \alpha(\theta_2 - \theta_1)]$$
- and $V_2 = V_1 [1 + \gamma(\theta_2 - \theta_1)]$
- i. Define the α and γ used in above expressions. [1]
- ii. Show that $\alpha = \frac{\gamma}{3}$ [2]
- b. Two bodies made of the same material have the same external dimensions and appearance, but one is solid and the other is hollow as shown in figure.



- i. Will their cubical expansivity be the same or different? Give justification of your answer. [1]
- ii. When they are heated to the same temperature range, is overall volume expansion the same or different? [1]
- c. The volume of a metal sphere is increased by 1% of its original volume, when it is heated from 20°C to 220°C . Calculate the coefficient of linear, superficial and cubical expansion of the metal. [3]

22. a. Complete the ray diagram to show total deviation produced by Prism. [1]



- b. Derive the relation, $\delta = i + e - A$, where symbols have their usual meaning. [2]
- c. Sketch the graph to explain the variation of angle of deviation with angle of incidence. [2]
- d. A glass prism of angle 72° and index of refraction 1.66 is immersed in a liquid of refractive index 1.33. What is the angle of minimum deviation for a parallel beam of light passing through the prism? [3]

☞ - The End - ☞